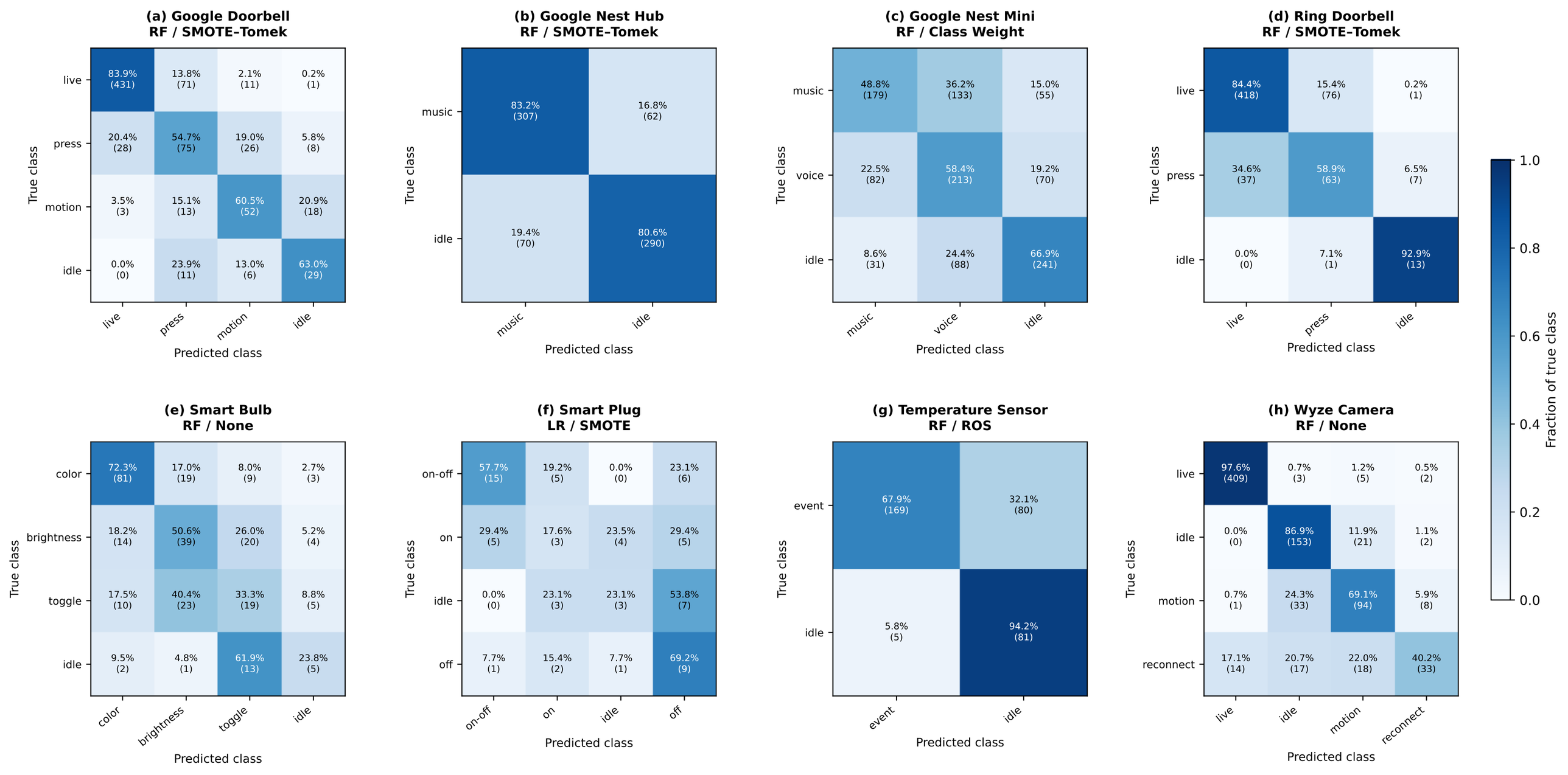
(a) Google Doorbell
RF / SMOTE–Tomek
True class
live
press
motion
idle
83.9% (431)
13.8% (71)
2.1% (11)
0.2% (1)
20.4% (28)
54.7% (75)
19.0% (26)
5.8% (8)
3.5% (3)
15.1% (13)
60.5% (52)
20.9% (18)
0.0% (0)
23.9% (11)
13.0% (6)
63.0% (29)
Predicted class
(b) Google Nest Hub
RF / SMOTE–Tomek
music
idle
83.2% (307)
16.8% (62)
19.4% (70)
80.6% (290)
(c) Google Nest Mini
RF / Class Weight
voice
48.8% (179)
36.2% (133)
15.0% (55)
22.5% (82)
58.4% (213)
19.2% (70)
8.6% (31)
24.4% (88)
66.9% (241)
(d) Ring Doorbell
RF / SMOTE–Tomek
84.4% (418)
15.4% (76)
0.2% (1)
34.6% (37)
58.9% (63)
6.5% (7)
0.0% (0)
7.1% (1)
92.9% (13)
Fraction of true class
1.0
0.8
0.6
0.4
0.2
0.0
(e) Smart Bulb
RF / None
color
brightness
toggle
72.3% (81)
17.0% (19)
8.0% (9)
2.7% (3)
18.2% (14)
50.6% (39)
26.0% (20)
5.2% (4)
17.5% (10)
40.4% (23)
33.3% (19)
8.8% (5)
9.5% (2)
4.8% (1)
61.9% (13)
23.8% (5)
(f) Smart Plug
LR / SMOTE
on-off
on
off
57.7% (15)
19.2% (5)
0.0% (0)
23.1% (6)
29.4% (5)
17.6% (3)
23.5% (4)
29.4% (5)
0.0% (0)
23.1% (3)
23.1% (3)
53.8% (7)
7.7% (1)
15.4% (2)
7.7% (1)
69.2% (9)
(g) Temperature Sensor
RF / ROS
event
67.9% (169)
32.1% (80)
5.8% (5)
94.2% (81)
(h) Wyze Camera
RF / None
reconnect
97.6% (409)
0.7% (3)
1.2% (5)
0.5% (2)
0.0% (0)
86.9% (153)
11.9% (21)
1.1% (2)
0.7% (1)
24.3% (33)
69.1% (94)
5.9% (8)
17.1% (14)
20.7% (17)
22.0% (18)
40.2% (33)